



AMIT KUMAR

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[LinkedIn](#) | [GitHub](#)

CAREER OBJECTIVE

Aspiring Electronics and Communication Engineering student with a strong foundation in Analog VLSI, layout design and circuit analysis. I am eager to apply my skills and contribute to innovative and efficient solutions that support the success of the organization.

EDUCATION

KLE TECHNOLOGICAL UNIVERSITY

B.E. Electronics & Communication Engineering

CGPA: 8.43 (up to 6th semester)

Hubballi, Karnataka

2022 - 2026

Shri Abhay Nobles Sr. Sec. School

Senior Secondary (Rajasthan Board of Secondary Education)

Percentage: 84.60%

Takhatgarh, Pali, Rajasthan

2022

Shri Abhay Nobles Sr. Sec. School

Secondary (Rajasthan Board of Secondary Education)

Percentage: 75.17%

Takhatgarh, Pali, Rajasthan

2020

PROJECTS

Design and Layout of a 7-Pack Operational Amplifier

- Designed a 7-pack operational amplifier with 59 dB gain, 56° phase margin, operating at a 1.8V power supply while achieving a high unity gain bandwidth (UGB) of 33MHz and ensuring optimal performance.
- Implemented a two-stage topology with Miller compensation to improve frequency response and ensure reliable operation.
- The op-amp is designed using UMC 180nm technology, and design rule checks (DRC) along with layout-versus-schematic (LVS) validation are performed to ensure layout correctness.

Design and Analysis of DC-DC Buck Converter

- Designed a DC-DC Buck Converter to convert an input voltage of 3.3V to a stable output voltage of 1.8V while supporting a load current of 100mA.
- Developed a Closed-Loop Control System utilizing Pulse Width Modulation (PWM) to regulate the output voltage, ensuring accurate and stable performance.
- Utilized Cadence tools with the UMC 330nm technology node for circuit design, schematic capture, simulation and verification.

Surface Waste Pollutants Classification using CNN & YOLOv8m

- Designed and compared a classification and detection framework for surface water pollutant using CNN and YOLOv8m for performance comparison across five waste categories.
- Used GAN-generated images to enhance dataset diversity.
- Achieved improved classification accuracy and enhanced real-time object detection through model optimization and inference benchmarking.

SKILLS

- Programming Languages:** C, C++, Embedded C, Python
- Core Subjects:** Analog Electronic Circuits, Circuit Analysis, Linear Integrated Circuits, CMOS VLSI, Control Systems
- Tools:** Cadence Virtuoso, LT spice, GitHub, MATLAB and Simulink
- Knowledge:** Device Matching, Guard ring, Floor planning and Routing
- Operating System:** Windows, Linux.
- Soft Skills:** Problem-solving, Leadership, Team Work, Adaptability, Active Listening
- Areas of Interest:** Analog Design, Analog Layout
- Hobbies:** Playing badminton, Listening to music and Podcast

PUBLICATIONS

Paper: 'A Detection Framework for Floating Pollutants on Water Bodies Using Object Detection Techniques'
Presented at **ICDSA 2025**, Malaviya National Institute of Technology (MNIT), Rajasthan, July 16-17, 2025.

CERTIFICATIONS

- Cadence** Analog IC Design Flow
- MATLAB** - Matlab Onramp and Simulink Onramp